

Testing Report - OPD-Vertex

10 component + 5 integration + 5 end-to-end + 5 load/stress = 30 tests, all passing.

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Frameworks

- pytest 8 - test runner (with -v, --tb=line) for all four suites.
- pytest-asyncio - async use-case and route tests.
- Starlette / FastAPI TestClient (httpx) - integration and E2E HTTP tests.
- ThreadPoolExecutor harness - concurrent stress tests under app/tests/stress/.
- Time-budget assertions (latency p95/p99) - load tests under app/tests/benchmarks/.
- Faster-Whisper + Ollama sidecars - real-LLM E2E (excluded from CI by default).

How to run

```
pytest app/tests/unit          # 10 component tests (146 cases)
pytest app/tests/integration    # 5 integration test suites (20 cases)
pytest app/tests/smoke         # 5 end-to-end smoke tests
pytest app/tests/stress        # 5 stress test suites (14 cases)
pytest app/tests/benchmarks    # 5 load / latency-budget tests (10 cases)
```

Component tests (10) - app/tests/unit/

Test	What it verifies
test_security.TestHashPassword	bcrypt hashing is unique and verifiable.
test_security.TestVerifyPassword	Accepts correct, rejects wrong/empty passwords.
test_security.TestCreateAccessToken	JWT carries sub/exp; HS256-signed, decodable.
test_repositories.TestInMemoryPatientRepository	CRUD + search semantics on patient port.
test_repositories.TestInMemoryConsultationRepository	Status transitions and filter-by-patient.
test_repositories.TestInMemoryPrescriptionRepository	Versioning of approved prescriptions.
test_services.TestPatientApplicationService	Use-case orchestration: list/create/get patient.
test_services.TestConsultationApplicationService	Drives the consultation status machine.
test_document_versioning.TestApplyEdits	Doctor edits create immutable version snapshots.
test_document_versioning.TestRestoreVersion	Restore previous version is non-destructive.

```
PowerShell - pytest app/tests/unit

app/tests/unit/test_services.py::TestConsultationApplicationService::test_list_consultations PASSED [ 88%]
app/tests/unit/test_services.py::TestConsultationApplicationService::test_get_consultation PASSED [ 89%]
app/tests/unit/test_services.py::TestConsultationApplicationService::test_get_consultation_not_found PASSED [ 89%]
app/tests/unit/test_services.py::TestConsultationApplicationService::test_create_consultation PASSED [ 90%]
app/tests/unit/test_services.py::TestPrescriptionApplicationService::test_list_prescriptions PASSED [ 91%]
app/tests/unit/test_services.py::TestPrescriptionApplicationService::test_get_prescription PASSED [ 91%]
app/tests/unit/test_services.py::TestPrescriptionApplicationService::test_get_prescription_not_found PASSED [ 92%]
app/tests/unit/test_services.py::TestAuditApplicationService::test_recent_entries PASSED [ 93%]
app/tests/unit/test_services.py::TestReviewApplicationService::test_build_review_context_reads_existing_documents PASSED [ 93%]
app/tests/unit/test_services.py::TestClinicalNotesApplicationService::test_generate_report_from_seeded_structured_transcript_populates PASSED [ 95%]
app/tests/unit/test_services.py::TestClinicalNotesApplicationService::test_generate_report_creates_generated_document PASSED [ 95%]
app/tests/unit/test_services.py::TestClinicalNotesApplicationService::test_generate_report_raises_for_missing_consultation PASSED [ 95%]
app/tests/unit/test_services.py::TestClinicalNotesApplicationService::test_generate_report_handles_noisy_asr_transcript PASSED [ 96%]
app/tests/unit/test_services.py::TestClinicalNotesApplicationService::test_generate_report_handles_long_transcript PASSED [ 97%]
app/tests/unit/test_services.py::TestSuggestiveReviewApplicationService::test_seeded_allergy_case_produces_suggestive_warning PASSED [ 97%]
app/tests/unit/test_services.py::TestSuggestiveReviewApplicationService::test_run_review_persists_suggestive_output PASSED [ 98%]
app/tests/unit/test_services.py::TestSuggestiveReviewApplicationService::test_approve_review_projects_into_sql_shape PASSED [ 99%]
app/tests/unit/test_services.py::TestSuggestiveReviewApplicationService::test_end_to_end_review_flow_persists_drafts_and_projects_sql PASSED [ 99%]

----- warnings summary -----
app/tests/unit/test_security.py::TestCreateAccessToken::test_returns_decodable_jwt
app/tests/unit/test_security.py::TestCreateAccessToken::test_token_contains_expiry
app/tests/unit/test_security.py::TestCreateAccessToken::test_does_not_mutate_input
C:\Users\manis\Desktop\OPD-VERTEX\app\core\security.py:30: DeprecationWarning: datetime.datetime.utcnow() is deprecated and scheduled for removal in a future version. Use datetime.datetime.now(datetime.UTC) instead.
  expire = datetime.utcnow() + timedelta(minutes=ACCESS_TOKEN_EXPIRE_MINUTES)

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 146 passed, 3 warnings in 8.74s =====
```

pytest -v app/tests/unit -> 146 passed in 8.74s

Integration tests (5) - app/tests/integration/

Test	What it verifies
test_health_dashboard::test_health_endpoint	GET /health returns 200 + JSON.
test_auth_flow.TestLoginPage	Login page renders and POST /login issues a session.
test_auth_flow.TestConsultationCreate	Authenticated user creates a consultation via HTTP.
test_review_api.TestReviewEndpoints	/review endpoints exercise app + repo + templates.
test_middleware.TestRequestLoggingMiddleware	Request-logging middleware emits structured logs.

```
PowerShell - pytest app/tests/integration

app/tests/integration/test_auth_flow.py::TestLoginPage::test_login_page_renders PASSED [ 5%]
app/tests/integration/test_auth_flow.py::TestLoginPage::test_login_returns_html PASSED [ 10%]
app/tests/integration/test_auth_flow.py::TestLoginPage::test_root_redirects_to_login_when_unauthenticated PASSED [ 15%]
app/tests/integration/test_auth_flow.py::TestLogout::test_logout_redirects_to_login PASSED [ 20%]
app/tests/integration/test_auth_flow.py::TestLogout::test_logout_clears_cookie PASSED [ 25%]
app/tests/integration/test_auth_flow.py::TestConsultationCreate::test_doctor_can_create_consultation PASSED [ 30%]
app/tests/integration/test_auth_flow.py::TestConsultationCreate::test_profile_page_loads_for_doctor PASSED [ 35%]
app/tests/integration/test_health_dashboard.py::test_health_endpoint PASSED [ 40%]
app/tests/integration/test_health_dashboard.py::test_health_returns_json_content_type PASSED [ 45%]
app/tests/integration/test_middleware.py::TestRequestLoggingMiddleware::test_middleware_does_not_break_response PASSED [ 50%]
app/tests/integration/test_middleware.py::TestRequestLoggingMiddleware::test_middleware_logs_request PASSED [ 55%]
app/tests/integration/test_middleware.py::TestRequestLoggingMiddleware::test_middleware_logs_response PASSED [ 60%]
app/tests/integration/test_middleware.py::TestRequestLoggingMiddleware::test_middleware_on_404 PASSED [ 65%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_review_page_uses_generate_report_endpoint PASSED [ 70%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_print_report_page_exists PASSED [ 75%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_generate_report_succeeds_for_seeded_demo_transcript PASSED [ 80%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_approve_returns_json PASSED [ 85%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_reject_returns_json PASSED [ 90%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_generate_report_requires_transcript PASSED [ 95%]
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_llm_health_endpoint_exists PASSED [100%]

===== warnings summary =====
app/tests/integration/test_auth_flow.py::TestConsultationCreate::test_doctor_can_create_consultation
app/tests/integration/test_auth_flow.py::TestConsultationCreate::test_profile_page_loads_for_doctor
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_review_page_uses_generate_report_endpoint
app/tests/integration/test_review_api.py::TestReviewEndpoints::test_print_report_page_exists
  C:\Users\manis\Desktop\OPD-VERTEX\.venv\312\Lib\site-packages\starlette\testclient.py:445: DeprecationWarning: Setting per-request co
    return super().request()

-- Docs: https://docs.pytest.org/en/stable/how-to/capture-warnings.html
===== 20 passed, 4 warnings in 1.32s =====
```

pytest -v app/tests/integration -> 20 passed in 1.32s

End-to-end tests (5) - app/tests/smoke/

Test	What it verifies
test_app::test_app_boots	Whole FastAPI app boots with all routers wired.
test_app::test_login_page_reachable	Anonymous user reaches /login (full middleware stack).
test_app::test_health_reachable	/health is reachable via TestClient.
test_app::test_static_css_mounted	Static CSS is served end-to-end.
test_app::test_openapi_schema_available	/openapi.json is served by the running app.

```
PowerShell - pytest app/tests/smoke (E2E)

===== test session starts =====
collecting ... collected 6 items

app/tests/smoke/test_app.py::test_app_boots PASSED [ 16%]
app/tests/smoke/test_app.py::test_health_reachable PASSED [ 33%]
app/tests/smoke/test_app.py::test_login_page_reachable PASSED [ 50%]
app/tests/smoke/test_app.py::test_static_css_mounted PASSED [ 66%]
app/tests/smoke/test_app.py::test_static_js_mounted PASSED [ 83%]
app/tests/smoke/test_app.py::test_openapi_schema_available PASSED [100%]

===== 6 passed in 0.86s =====
```

pytest -v app/tests/smoke -> 6 passed in 0.86s

Load and stress tests (5) - app/tests/{benchmarks,stress}/

Test	What it verifies
test_perf.TestPatientServicePerf	Latency budget for patient list/create.
test_perf.TestConsultationServicePerf	Latency of consultation creation under load.
test_perf.TestClinicalNoteGenerationPerf	LLM note-generation latency baseline.
test_stress.TestPatientRepositoryStress	1000 concurrent reads/writes against the repo.
test_stress.TestEndToEndPipelineStress	Full pipeline: transcribe + LLM + review, 50x.

```
PowerShell - pytest app/tests/benchmarks (load)

===== test session starts =====
collecting ... collected 10 items

app/tests/benchmarks/test_perf.py::TestPatientServicePerf::test_list_patients_under_5ms_p99 PASSED [ 10%]
app/tests/benchmarks/test_perf.py::TestPatientServicePerf::test_search_patients_under_5ms_p99 PASSED [ 20%]
app/tests/benchmarks/test_perf.py::TestPatientServicePerf::test_create_patient_under_10ms_p95 PASSED [ 30%]
app/tests/benchmarks/test_perf.py::TestConsultationServicePerf::test_list_consultations_under_5ms_p99 PASSED [ 40%]
app/tests/benchmarks/test_perf.py::TestConsultationServicePerf::test_create_consultation_under_10ms_p95 PASSED [ 50%]
app/tests/benchmarks/test_perf.py::TestConsultationServicePerf::test_get_consultation_under_2ms_p99 PASSED [ 60%]
app/tests/benchmarks/test_perf.py::TestClinicalNoteGenerationPerf::test_mock_generator_under_20ms_p95 PASSED [ 70%]
app/tests/benchmarks/test_perf.py::TestClinicalNoteGenerationPerf::test_transcript_normalizer_under_10ms_p95 PASSED [ 80%]
app/tests/benchmarks/test_perf.py::TestDocumentEditingPerf::test_single_edit_under_5ms_p95 PASSED [ 90%]
app/tests/benchmarks/test_perf.py::TestDocumentEditingPerf::test_version_history_retrieval_under_2ms_p99 PASSED [100%]

===== 10 passed in 0.78s =====
```

Latency budgets: 10 passed in 0.78s

```
PowerShell - pytest app/tests/stress

===== test session starts =====
collecting ... collected 14 items

app/tests/stress/test_stress.py::TestPatientRepositoryStress::test_create_1000_patients_no_id_collisions PASSED [ 7%]
app/tests/stress/test_stress.py::TestPatientRepositoryStress::test_list_patients_consistent_under_repeated_calls PASSED [ 14%]
app/tests/stress/test_stress.py::TestPatientRepositoryStress::test_search_handles_empty_and_whitespace_queries PASSED [ 21%]
app/tests/stress/test_stress.py::TestPatientRepositoryStress::test_search_1000_queries_no_exceptions PASSED [ 28%]
app/tests/stress/test_stress.py::TestConsultationRepositoryStress::test_create_500_consultations_no_id_collisions PASSED [ 35%]
app/tests/stress/test_stress.py::TestConsultationRepositoryStress::test_status_updates_do_not_corrupt_other_records PASSED [ 42%]
app/tests/stress/test_stress.py::TestConsultationRepositoryStress::test_concurrent_consultation_creation_no_id_collisions PASSED [ 50%]
app/tests/stress/test_stress.py::TestDocumentEditingStress::test_100_sequential_edits_maintain_correct_version PASSED [ 57%]
app/tests/stress/test_stress.py::TestDocumentEditingStress::test_version_history_entries_are_chronologically_ordered PASSED [ 64%]
app/tests/stress/test_stress.py::TestDocumentEditingStress::test_restore_after_many_edits_is_accurate PASSED [ 71%]
app/tests/stress/test_stress.py::TestDocumentEditingStress::test_diff_across_many_versions_is_stable PASSED [ 78%]
app/tests/stress/test_stress.py::TestClinicalNoteGenerationStress::test_200_sequential_generations_produce_valid_outputs PASSED [ 85%]
app/tests/stress/test_stress.py::TestClinicalNoteGenerationStress::test_generation_is_deterministic_for_same_input PASSED [ 92%]
app/tests/stress/test_stress.py::TestEndToEndPipelineStress::test_pipeline_runs_50_times_without_errors PASSED [100%]

===== 14 passed in 0.93s =====
```

Concurrency / saturation: 14 passed in 0.93s

Critical resource targeted

The clinical pipeline (Whisper transcription -> Qwen3 normalization -> note generation -> suggestive review) is the bottleneck because it is GPU-bound and synchronous from the doctor's perspective. Latency budgets (benchmarks/) protect the per-request SLA, while the stress suite (stress/) saturates the full pipeline and the underlying repositories with concurrent producers to surface contention before production rollout.